

Program : Diploma in Cloud Computing and Big Data	
Course Code : 6279A	Course Title: Block Chain Lab
Semester : 6	Credits: 1.5
Course Category: Programme Elective Course	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Provide a conceptual understanding of how block chain technology works in real-world application
- Familiarize with the block Chain operations in practical implementation of solutions using block Chain technology.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basics in Python	1008	Introduction to IT Systems Lab	I
Demonstrate Python Programs	3279	Programming in python Lab	III

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Implement blockchain fundamentals using Python programming	9	Applying
CO2	Perform the validity check and display of implemented blockchains	9	Applying
CO3	Experiment with transactions in Metamask	8	Applying
CO4	Deploy a smart contract	13	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			3
CO2	3			3			3
CO3	3			3			3
CO4	3			3			3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Implement blockchain fundamentals using Python programming		
M1.01	Create blocks	3	Applying
M1.02	Formulate blockchains	3	Applying
M1.03	Perform Hash Verification	3	Applying
CO2	Perform the validity check and display of implemented blockchains		
M2.01	Experiment mining on the blocks	3	Applying
M2.02	Perform Display the block	3	Applying
M2.03	Implement validity Checking of blockchain	3	Applying
	Lab Test – I	3	Applying
CO3	Experiment with transactions in Metamask		
M3.01	Demonstrate Metamask	2	Applying
M3.02	Create several Ethereum accounts	2	Applying

M3.03	Demonstrate how to Deposit Ether in Metamask	2	Applying
M3.04	Develop some transactions between the accounts	2	Applying
CO4	Deploy a smart contract		
M4.01	Demonstrate REMIX	3	Applying
M4.02	Experiment with default modules	3	Applying
M4.03	Demonstrate default solidity modules	4	Applying
M4.04	Deploy a sample smart contract in Solidity	3	Applying
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
T1	David Metcalf, Max Hooper, and Vikram Dhillon “Blockchain Enabled Applications: Understand the Blockchain Ecosystem and How to Make it Work for You” -Createspace Independent Pub (2017)
T2	Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder, “Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction”, Princeton University Press, 2016.

Online Resources

Sl.No	Website Link
1	https://www.geeksforgeeks.org/create-simple-blockchain-using-python/
2	https://kbaiitmk.medium.com/
3	https://learn.kba.ai/
4	https://docs.metamask.io/guide/#why-metamask .
5	https://remix-ide.readthedocs.io/